HACKATHON
PROBLEM STATEMENT

AI-Powered Personal Knowledge & Career Intelligence Platform

From fragmented learning resources to structured knowledge, measurable skills, and career readiness.

1. The Problem

In today's information-driven world, individuals have access to an overwhelming amount of learning material in the form of books, research papers, journals, PDFs, presentations, online courses, and other digital resources. However, accessing information is not the same as understanding, retaining, or applying it.

Most learning platforms offer fragmented, content-centric experiences. They help users consume information but fail to build a structured, interconnected understanding of concepts over time. Learners often struggle to organize knowledge, identify gaps in their understanding, revise effectively, and translate what they have learned into practical skills.

The challenge becomes even more significant when individuals need to apply their knowledge in professional settings, particularly during job applications and interviews. Existing recruitment tools primarily focus on matching resumes with job descriptions through keywords and predefined criteria, rather than evaluating a candidate's actual knowledge, competencies, and readiness for a role.

There is a need for an intelligent platform that bridges the gap between information consumption, knowledge development, skill assessment, and career readiness.

2. Proposed Solution

Build an AI-powered Personal Knowledge and Career Intelligence Platform that transforms a user's learning resources into a structured, interactive knowledge base and uses that knowledge to support continuous learning, skill development, and interview preparation.

Phase I: AI-Powered Personal Knowledge Management & Learning System

Develop an intelligent learning ecosystem that converts unstructured learning materials into a structured, interconnected, and continuously evolving knowledge base.

- **Multi-format content ingestion:** Allow users to upload PDFs, books, eBooks, research papers, journals, presentations, documents, and other learning resources.
- **Intelligent knowledge extraction:** Use AI to identify key concepts, topics, definitions, frameworks, relationships, and insights from uploaded materials.

- **Dynamic knowledge graph:** Automatically build a visual, interconnected knowledge graph that maps relationships between concepts and evolves as users add more learning resources.
- **Personalized learning:** Enable users to explore topics, understand complex concepts, ask contextual questions, and learn through AI-powered explanations.
- **Knowledge assessment:** Generate quizzes, flashcards, practice questions, and assessments to evaluate comprehension and retention.
- **Continuous knowledge development:** Identify knowledge gaps, recommend relevant learning materials, and help users progressively expand their personal knowledge base.

Expected outcome: A centralized, intelligent knowledge ecosystem that helps individuals move beyond consuming information to building, retaining, and applying knowledge.

Phase II: AI-Powered Career Intelligence & Interview Preparation System

Extend the knowledge platform into an intelligent career preparation and assessment system that connects an individual's knowledge, experience, and skills with real-world job requirements.

- **Resume and Job Description Analysis:** Allow users to upload their resumes and target job descriptions. Use AI to analyze both and identify alignment, missing competencies, experience gaps, and relevant keywords.
- **Intelligent Resume Optimization:** Recommend improvements to resume structure, content, achievements, keywords, and positioning based on the target role.
- **Personalized Skill Gap Analysis:** Identify technical, functional, and behavioral competencies required for the role that are missing or insufficiently demonstrated in the candidate's profile.
- **Knowledge-Based Interview Preparation:** Leverage the user's existing knowledge base to create role-specific interview questions, case studies, assessments, and practical exercises.
- **AI Interview Mentor:** Simulate interviews, evaluate responses, provide feedback, and identify areas requiring improvement.
- **Personalized Learning Roadmap:** Recommend additional concepts, skills, and learning resources to bridge identified gaps and improve job readiness.
- **Continuous Readiness Assessment:** Track the user's progress and measure improvements in knowledge, competencies, and interview preparedness.

Expected outcome: An intelligent career companion that helps individuals understand where they stand, what they are missing, and what they need to learn to become competitive candidates for their target roles.

3. The Core Innovation

The real differentiation should not be another AI chatbot that summarizes PDFs or another ATS tool that matches keywords. It should be the connection between the two systems.

1. KNOWLEDGE INGESTION	Books · PDFs · Research Papers · PPTs · Documents
↓	
2. PERSONAL KNOWLEDGE GRAPH	Extract · Structure · Connect · Organize
↓	
3. LEARNING & ASSESSMENT	Learn · Revise · Practice · Assess
↓	
4. CAREER INTELLIGENCE	Resume Analysis · JD Matching · Skill Gap Analysis
↓	
5. INTERVIEW READINESS	Mock Interviews · Feedback · Personalized Learning Roadmap

Feedback loop: Insights from quizzes, assessments, and mock interviews continuously inform the user's learning priorities and help strengthen the knowledge graph.

4. Hackathon Challenge

How might we leverage Generative AI, Large Language Models, and Knowledge Graphs to build a personalized intelligence platform that transforms unstructured learning resources into structured knowledge, continuously evaluates an individual's competencies, and bridges the gap between what they know and what they need to know to achieve their academic and professional goals?

5. Vision

To build a unified intelligence platform that becomes an individual's lifelong learning and career companion.

Rather than maintaining separate tools for reading, learning, note-taking, assessments, resume optimization, and interview preparation, users should have one continuously evolving system that understands:

- What they know.
- What they don't know.
- How different concepts connect.
- What they need to learn next.
- How effectively they can apply their knowledge.
- How prepared they are for their desired career opportunities.

Positioning recommendation

Position Phase I as the core product and Phase II as its most powerful application. The strongest proposition is not an ATS with a learning feature. It is a Personal Knowledge Intelligence System that translates knowledge into measurable skills and career outcomes.